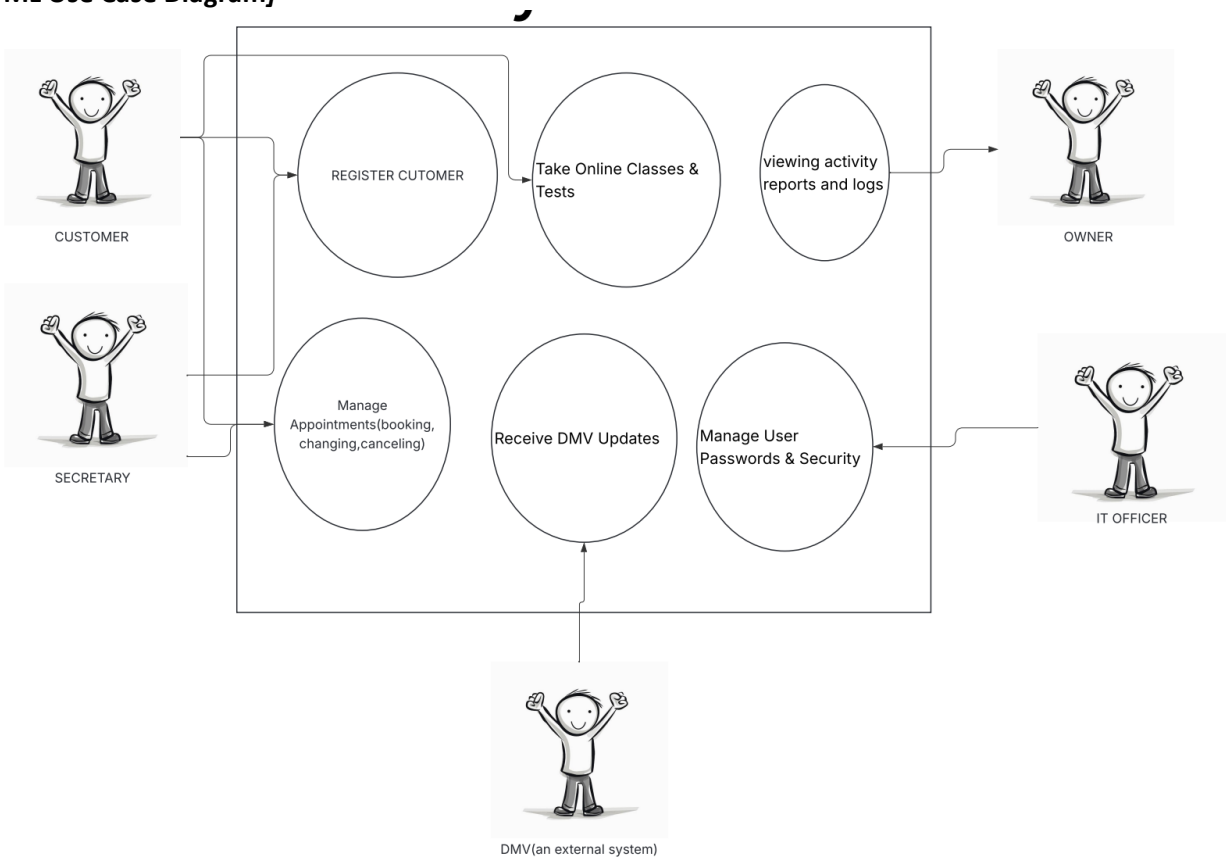
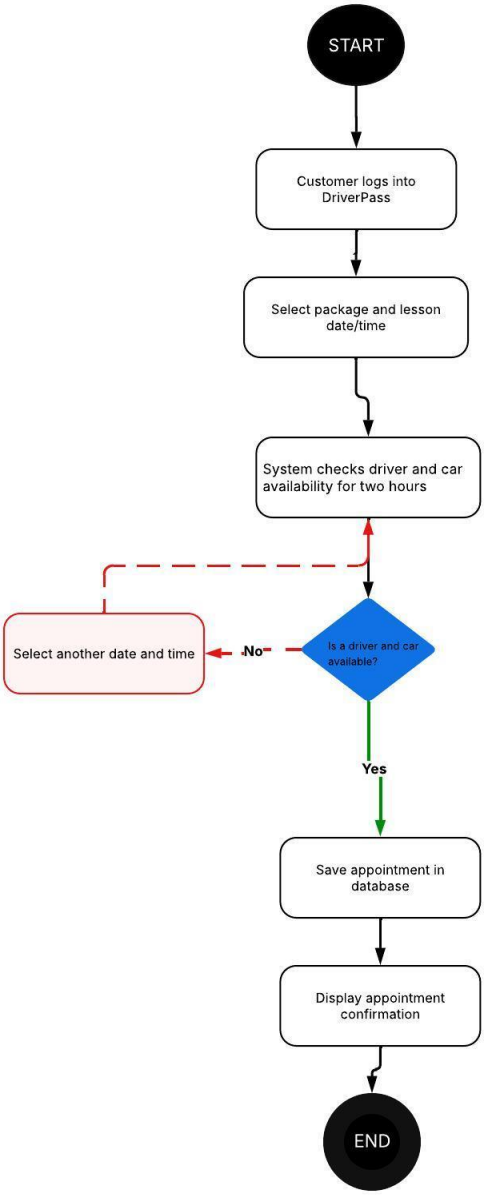
CS 255 System Design Document Template

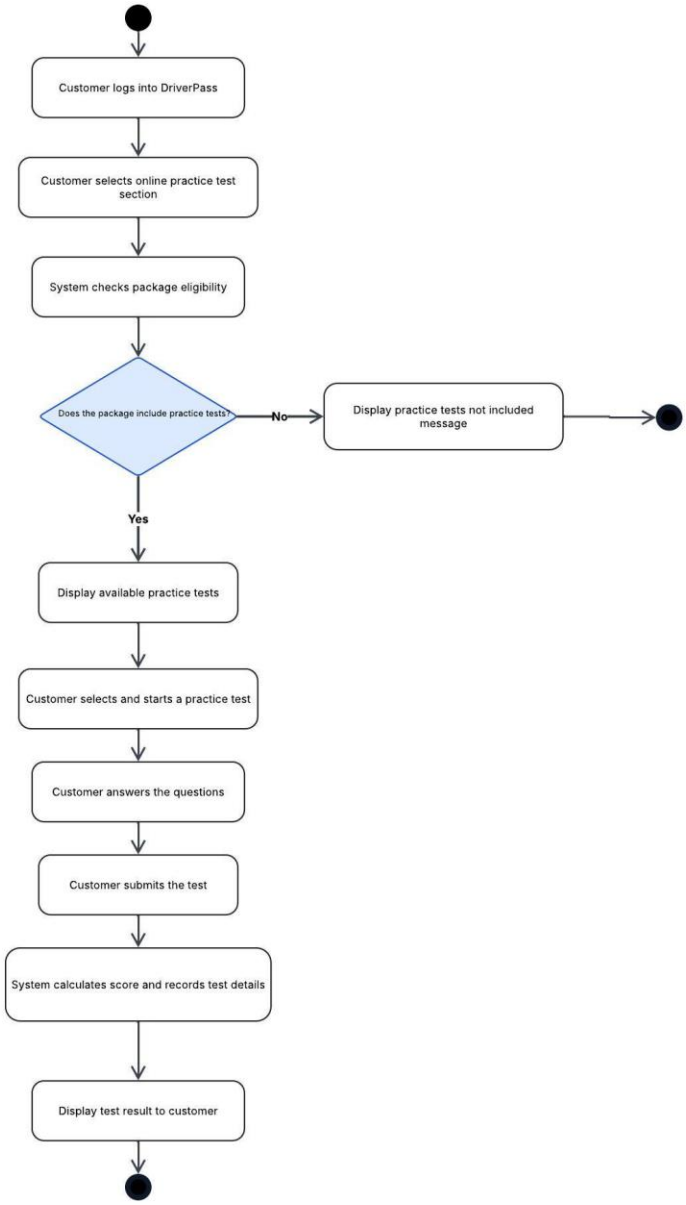
UML Diagrams

UML Use Case Diagram*]*

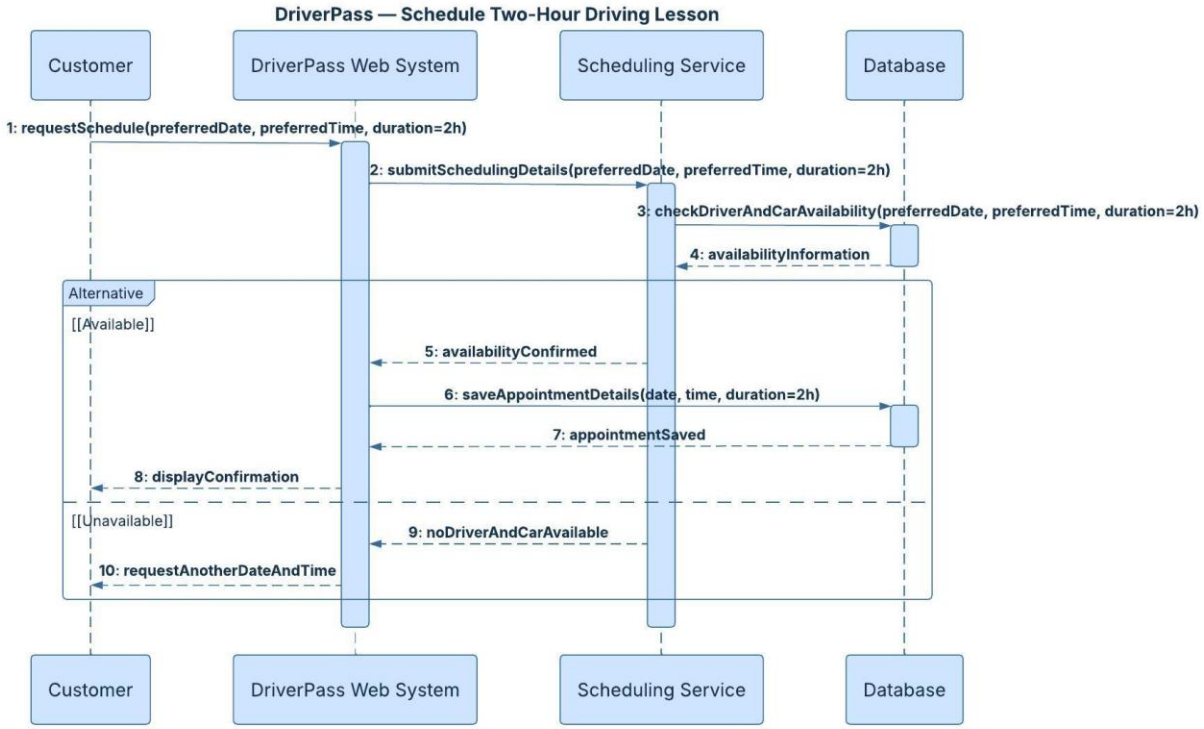


UML Activity Diagrams

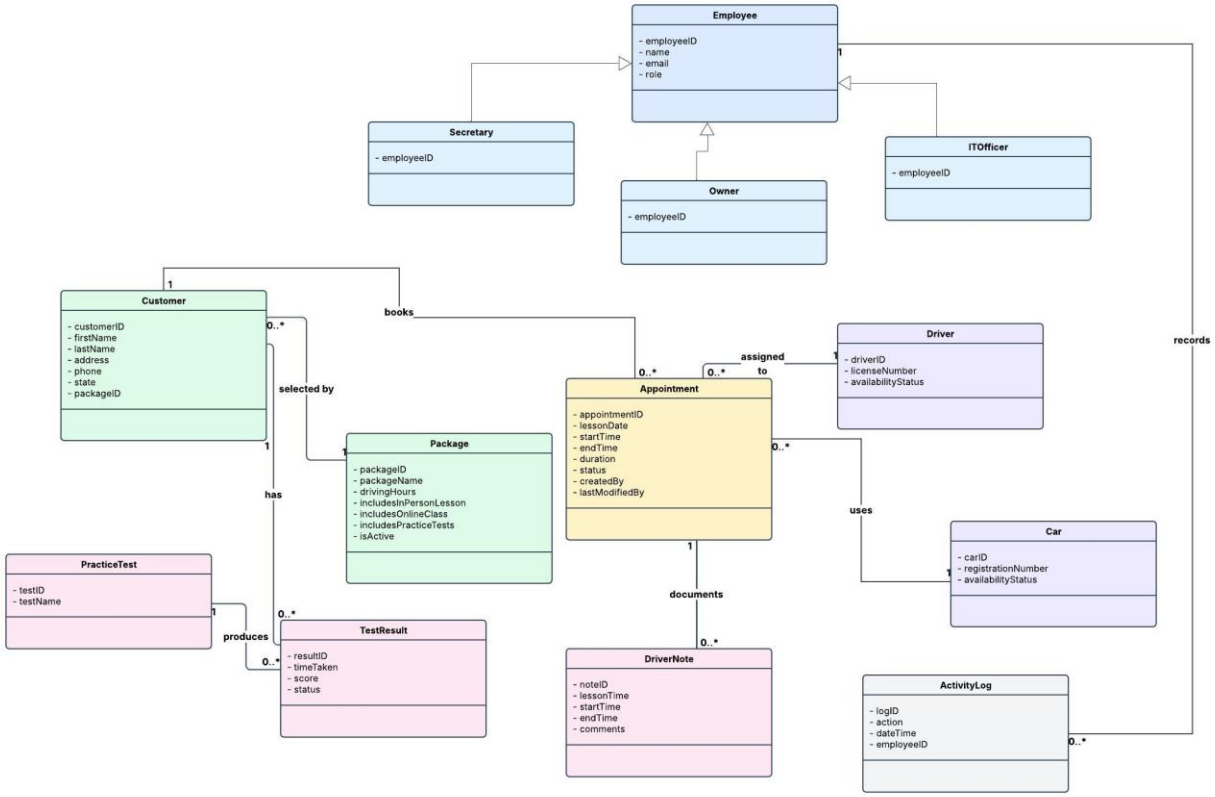

START
Customer logs into DriverPass
Select package and lesson date/time
System checks driver and car availability for two hours
Is a driver and car available?
No
Select another date and time
Yes
Save appointment in database
Display appointment confirmation
END



UML Sequence Diagram



UML Class Diagram



Technical Requirements

Southern New Hampshire University

Technical Requirements

Technical Requirements

Hardware

DriverPass will run online, so the company will need a server environment to host the application and its database. Using a cloud service would allow customers and employees to access the system without being in the same location. Customers can use their phones, tablets, or computers, while employees can use office computers to handle appointments and other daily tasks. The system should also be able to handle several people using it at once.

Software

The main application will be a website that users can open through a web browser. Customers should be able to register, choose a package, arrange driving lessons, and complete online practice tests if their package includes them. Employees will use the system to manage customer accounts, driving schedules, drivers, and vehicles.

A database will keep the information organized, including customer details, package options, appointments, and test results. Employees with the correct permissions should also be able to create reports and export them for use in Excel.

Security

Since DriverPass will handle personal and payment-related information, protecting customer data is important. Users should sign in with individual accounts, and the system should limit what each person can access based on their job. For example, the IT officer needs to manage accounts, while customers should be able to view their own information and appointments.

The website should use an encrypted connection, and passwords must be stored safely. It should also limit repeated failed login attempts and provide a password recovery option. Appointment changes should be recorded so the company can check who created, changed, or cancelled a booking. Payment processing should use a secure payment service, and the system should not retain card security codes.

Development Tools

Lucidchart will be used to prepare the UML diagrams for the design. During development, the programmers will need tools to write the application, manage the database, and test the software. Testing should cover common tasks such as booking a lesson, checking vehicle availability, changing an appointment, and saving practice test results.

Hosting and Maintenance

Because DriverPass needs its information to stay updated, the system will require an internet connection whenever users make changes. A cloud hosting service can provide the environment for the website and database, along with backup and maintenance services. The system should save updates correctly and respond without unnecessary delays.

Employees should be able to manage accounts and turn package options on or off when needed. However, adding major features or changing how packages work may require a programmer. DriverPass will also need to keep its practice test information current when the DMV provides updated material.